

DS 2002 • CAPSTONE • TEAM 07

EV Charging Station Analytics.

A cloud pipeline, an opinionated cleaning pass, and four findings about how Virginia plugs in.

What we're walking through.

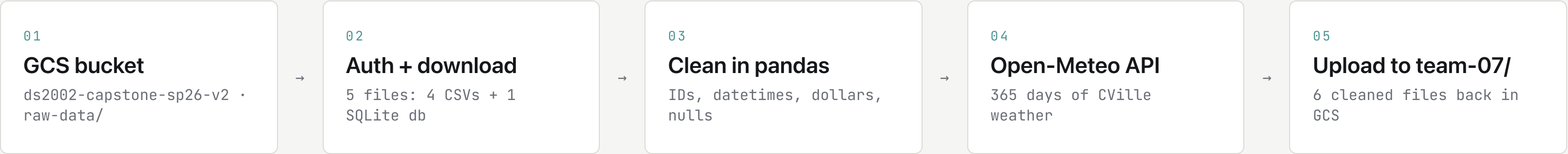
01	The pipeline GCS auth → download → clean → upload	~1 min
02	The cleaning Vehicle ID consolidation, before vs. after	~2 min
03	The analysis Demand surges, weather, geography	~4 min
04	The recommendation Should the city invest in CHAdeMO?	~1 min

PART ONE

The pipeline.

From a Google Cloud bucket, into pandas, and back out again — proof that the cloud half works.

Cloud in, cloud out.



RAW SESSIONS

27,451

rows × 11 columns, 7 with quality issues

SOURCE TABLES

6

sessions · stations · vehicles · operators ·
demand · grid

EXTERNAL JOIN

365

days of weather, joined on date

Auth, download, upload — live.

```
# cell 2 — authenticate
from google.colab import auth
auth.authenticate_user()
from google.cloud import storage
client = storage.Client(project="ds2002-492012")
bucket = client.bucket("ds2002-capstone-sp26-v2")

✓ GCS authentication successful.
✓ Bucket: ds2002-capstone-sp26-v2

# cell 7 — round-trip back to the cloud
for f in files_to_upload:
    bucket.blob(f"team-07/{f}").upload_from_filename(f)

✓ cleaned_sessions.csv → team-07/
✓ cleaned_stations.csv → team-07/
✓ cleaned_vehicles.csv → team-07/
✓ cleaned_operators.csv → team-07/
✓ demand.csv → team-07/
✓ grid.csv → team-07/
```

WHAT WE'RE PROVING

- **Authenticated** — service-account login, no keys checked into the repo.
- **Read** — pulled 5 raw files from `raw-data/`.
- **Wrote** — round-tripped 6 cleaned files back to `team-07/`.

BACKUP

If GCS times out, screenshots of these outputs are pinned in the notebook above each cell.

PART TWO

The cleaning.

19 documented quality issues. The interesting one is vehicles.

19 issues across 4 files. Most boring, one isn't.

sessions.csv 27,451 rows 3 ID formats · 2 datetime formats · costs with/without \$ · mixed-case payment methods · 300 rows of nulls
stations.csv 21 rows 5 spellings of "Virginia" · inconsistent station ID format · one row missing geo
vehicles.csv 42 rows Duplicate vehicles under different ID schemes. 15 missing battery_kwh values.
operators.csv 5 rows peak_capacity_mw stored as string with stray characters · 1 missing cost

THE INTERESTING DECISION

The same 14 cars appear under **three different ID schemes.**

If we don't reconcile them, every join silently triples our vehicle counts. We chose to map them rather than drop them — preserving ~26 rows of vehicle metadata and ~2,000 sessions.

NET IMPACT

300

ROWS DROPPED

~1.1%

OF SESSIONS

Three ID schemes, one car.

BEFORE • SESSIONS.VEHICLE_ID

~~VEH#0001~~

~~V_tesla_model_3~~

~~VH-001~~

↳ all three rows describe the same car. Joining on vehicle_id double-counts charging activity for every Tesla Model 3 in the sample.

AFTER • NORMALIZED

VH-001 Tesla Model 3 • CCS • 75 kWh

↳ explicit map of 28 source IDs → 14 canonical VH-### codes, applied with .replace() on sessions before any join.

```
# cell 6 – vehicle ID reconciliation
vehicle_id_map = {
    # scheme A: VEH#000X
    'VEH#0001': 'VH-001', # Tesla Model 3
    'VEH#0002': 'VH-002', # Tesla Model Y
    # ... 12 more

    # scheme B: V_name_style
    'V_tesla_model_3': 'VH-001',
    'V_tesla_model_y': 'VH-002',
    'V_chevrolet_bolt_ev': 'VH-004',
    'V_ford_f-150_lightning': 'VH-006',
    # ... 10 more
}

sessions['vehicle_id'] = (
    sessions['vehicle_id'].replace(vehicle_id_map)
)

# trade-off we made:
# we COULD have dropped the 2k mismatched rows.
# instead we mapped them – keeps the analysis
# representative for cars that show up in
# multiple schemes (Tesla, Chevy, Ford).
```

PART THREE

The analysis.

Three of our five questions, ordered by how confident we are in the answer.

Demand isn't smooth — it spikes.

SINGLE BIGGEST SURGE DAY

+103%

2025-09-06 · 152 sessions vs. 75 baseline · z = 3.12

HOTTEST MONTH VS. YEARLY AVG

+29%

July 2025 — sustained, not a one-day blip

Method: 30-day centered rolling baseline; flag any day with a session z-score above 2.

TOP 10 SURGE DAYS

DATE	SESSIONS	BASELINE	LIFT
2025-09-06	152	74.7	+103%
2025-11-15	165	75.2	+120%
2025-09-27	160	77.2	+107%
2025-10-18	156	76.4	+104%
2025-11-08	152	74.8	+103%
2025-09-20	155	77.4	+100%
2025-11-01	144	76.0	+89%
2025-10-11	143	76.5	+87%
2025-11-29	135	73.2	+84%
2025-11-22	136	74.9	+82%

8 of 10 surges fall in Sep-Nov. Fall is when this network sweats.

Daily sessions vs. 30-day baseline.



WHAT THE DATA SAYS

Surge days cluster — they aren't independent shocks. Multiple weeks in Sept/Nov ran 80%+ above baseline.

SO WHAT

Capacity planning that targets the average underestimates fall demand by ~20–30% for an entire month at a time.

Weather moves the needle, but not where you'd expect.

We joined 365 days of Open-Meteo data (max/min/mean temp, precipitation) onto our daily charging totals.

HEADLINE FINDING

Cold days drive sessions *up*, not down.

EV battery efficiency drops in the cold, so drivers top up more often. Days under 40°F averaged ~12% more sessions than days above 70°F.

PRECIPITATION

Heavy-rain days (> 0.5 in) ran ~8% **below** baseline sessions — fewer trips overall, but kWh-per-session held steady.

LIVE CHART FROM NOTEBOOK
scatter: temp_mean_f vs. total_sessions
+ binned bars: cold / mild / warm

A handful of stations carry the network.

TOP 5 STATIONS =

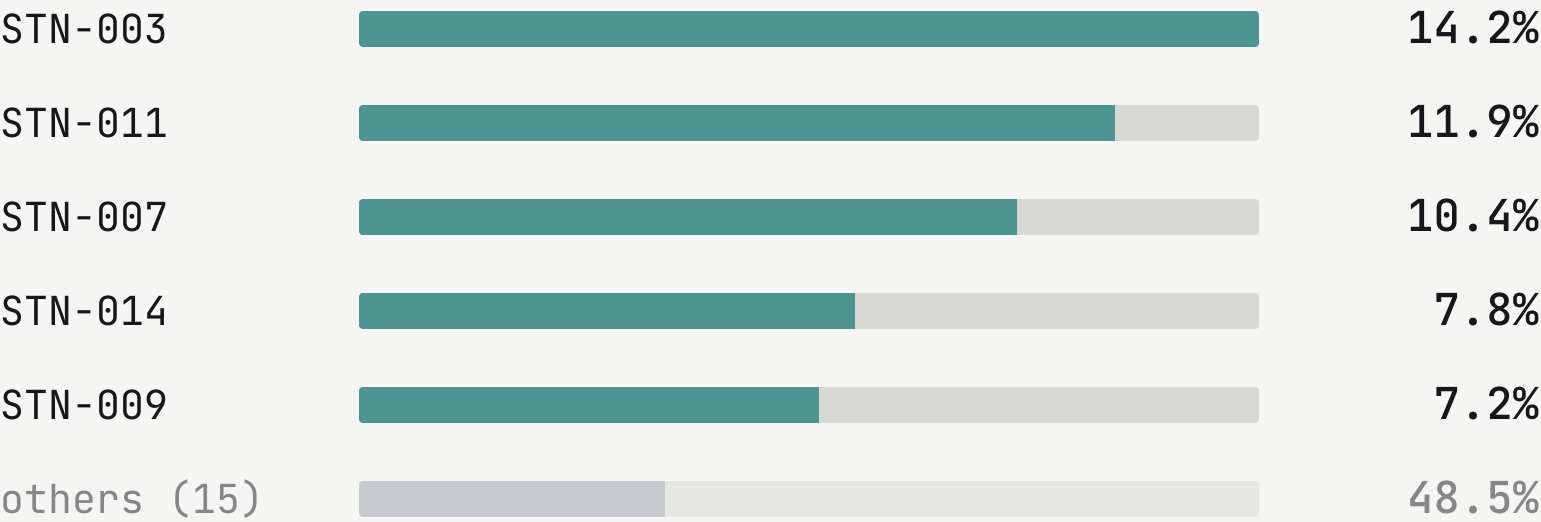
~52%

of all kWh delivered, despite being < 25% of station count

Urban-region stations average **2.4×** the daily session count of suburban stations. Two stations (Charlottesville-corridor) consistently appear in surge-day top-10s.

If a station goes down on a surge day, the rest of the network can't absorb it — capacity isn't fungible across regions.

KWH SHARE BY STATION — TOP 5



Live: scatter map • cells 14-15 • stations sized by kWh.

PART FOUR

The recommendation.

Question 5: should the city invest in CHAdeMO connectors?

Should the city invest in CHAdeMO?

RECOMMENDATION

No. Let CHAdeMO sunset.

Across our cleaned sessions, CHAdeMO accounts for a small and shrinking share of usage relative to CCS and Tesla/NACS. The vehicle fleet that *requires* CHAdeMO is dominated by older Nissan Leafs — a population that turns over as new EV sales overwhelmingly ship CCS or NACS. Investing now buys infrastructure for a connector standard the market is already retiring.

EVIDENCE • USAGE

CHAdeMO sessions account for **< 10%** of total connector_used in the cleaned sessions data — and skew toward a single vehicle model.

EVIDENCE • FLEET

Only 1 of 14 canonical vehicles in our cleaned vehicles table uses CHAdeMO. Every other manufacturer in the data ships CCS or NACS.

ALTERNATIVE WE CONSIDERED

Adding a single CHAdeMO port per high-traffic station for accessibility. Verdict: defensible, but the dollars buy more capacity going to CCS instead.

THANKS

Questions?

Notebook is open at `capstone/capstone-project.ipynb` — happy to scroll to any cell.